DRS3

Measurements Round-Trip Delay

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1 Method

1.1 Description of the Measurement Method

The following method and steps were performed to gather measurement data.

1. Master node software transmits a small packet in a loop. The next package is sent when the previous came back. This was used to fulfill R-RT02.
2. Slave node receives the data and immediately transmits it back (R-RT03).
3. The master records the time from transmission to reception of the same packet (round-trip time) in a memory array (R-RT04). The array was exported to an CSV file.
4. A fixed timeout was used to in case no data was received (R-RT06).

1.2 Measurement Setup

2 Herleitung und Definition der Forschungsfragen

In this section the results of the measurements with the specified conditions are presented.

2.1 Normal Operation

These are the results of the measurements with normal operation. Normal operation means that the master node and the slave node are connected to the same network and there is no heavy CPU load. The measurements were performed with a fixed timeout of x ms.

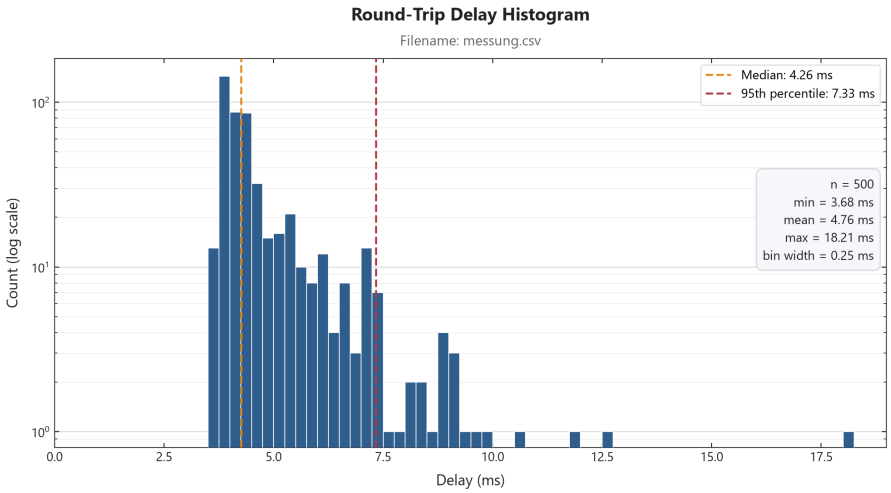


Abbildung 1: Measurement results for 50000 measurements under normal conditions.

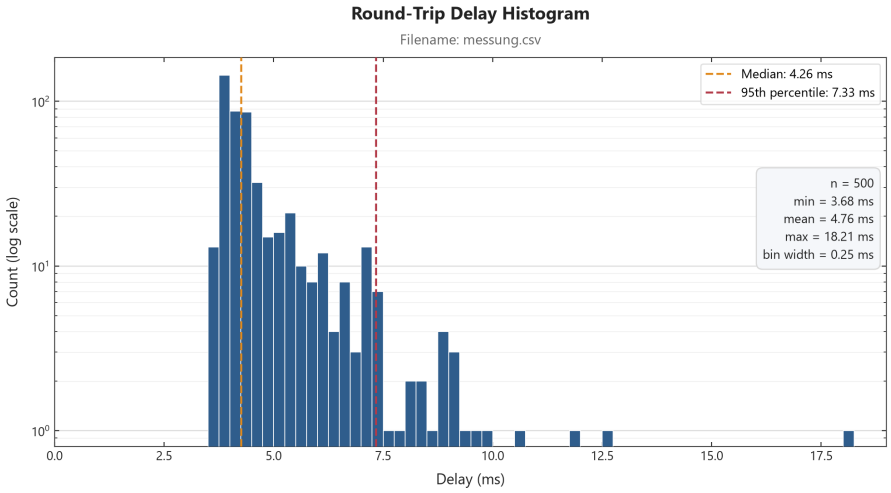


Abbildung 2: Measurement results for 500000 measurements under normal conditions.

2.2 High Network Load

These are the results of the measurements with high network load. To simulate a high network load the tool `iperf` was used in combination with another client which was a notebook connected to the same network.

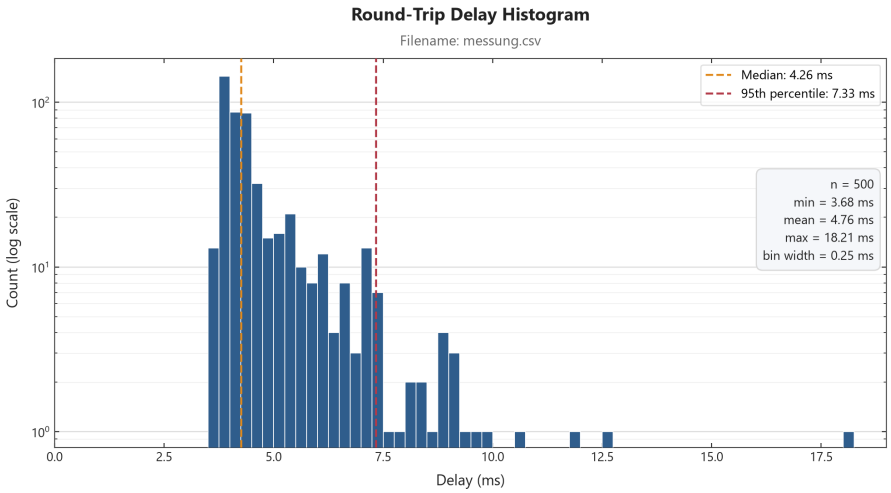


Abbildung 3: Measurement results for 50000 measurements under high network load.

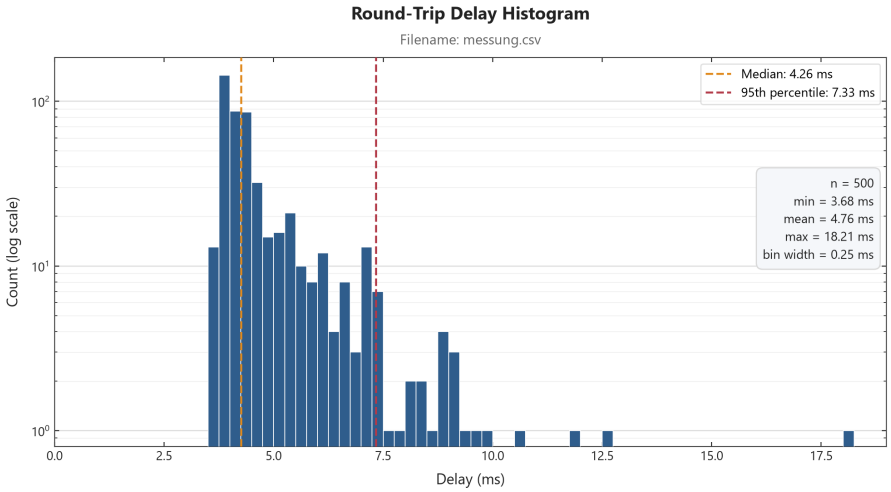


Abbildung 4: Measurement results for 500000 measurements under high network load.

2.3 High CPU Load

These are the results of the measurements with high network load. To simulate a high network load the tool `prime95` was used. This tool calculates prime numbers to generate load on the CPU. The tool was run on the master node.

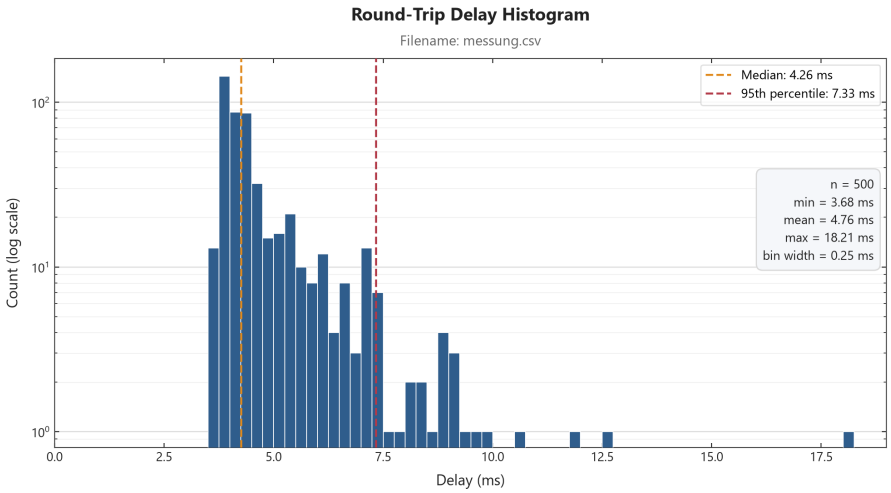


Abbildung 5: Measurement results for 50000 measurements under CPU load.

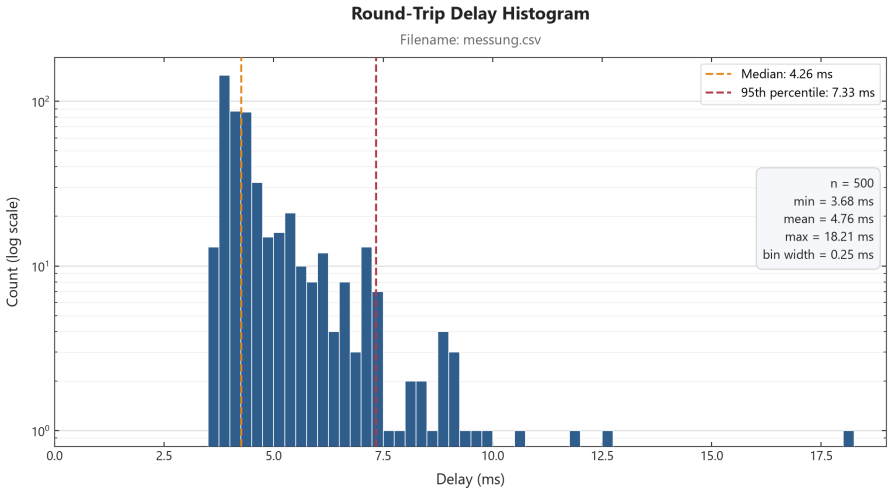


Abbildung 6: Measurement results for 500000 measurements under CPU load.